

Research on Optimization of Flight Scheduling Problem Based on the Combination of Ant Colony Optimization and Genetic Algorithm

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Abstract—As the flight scheduling has always been a key technology which reduces flight delays and costs. Rational and efficient flight sorting method not only can effectively improve the utilization of the airport, reduce flight delay, but also ensure flight safety and reduce the incidence of sudden accidents. Firstly, we studied the application of ant colony algorithm in flight sorting, and proposed a new flight sorting method which is combined with ant colony algorithm and genetic algorithm depending on the mutation characteristic of genetic algorithm; Secondly, the minimization objective model that the flight total delay was minimum was established based on the sorting method; Finally, the simulation experiment was done based on the model. The simulation results show that the proposed method can effectively reduce flight delay. Comparing with ant colony algorithm, the global optimal flight sequence can be found in a shorter time.

Keywords—flight scheduling; ant colony optimization; genetic algorithm; global optimal

In recent years, experts and scholars from home and abroad achieved ideal outcomes by using the ant colony algorithm and genetic algorithm to optimize the flight sequence. Studies have shown that optimal scheduling algorithm allows the system capacity increased by 15% in extreme cases, it is not only a tactical of air traffic automation that must be addressed, but also the capacity of the strategic of air traffic management systems that are necessary to assess[1]. However, a pure ant colony or a pure genetic algorithm has some shortcomings which are large computational overhead and large number of iterations to obtain the optimal solution and other shortcomings; Therefore, this article will combine ant colony algorithm and genetic algorithm and proposing ACO-GA algorithm. Meanwhile, the results of sorting through simulation experiments which use ACO-GA algorithm and FCFS algorithm, ant colony algorithm were compared, the results show that the ACO-GA flights sorting algorithm is superior to FCFS and ACO algorithm for optimization problems.

1 ACO Algorithm Overview

Ant colony algorithm (ACO) is a simulation evolutionary algorithm that inspired by the foraging behavior of ants in nature. Figure 1 shows the ants foraging routing simulations.

There are some advantages that ACO has, such as strong robustness, excellent distributed computer system, easy to combine with other methods, etc[2]. Ant colony algorithm is not easy to fall into local optimal solution, and can quickly get global optimal solution[2].

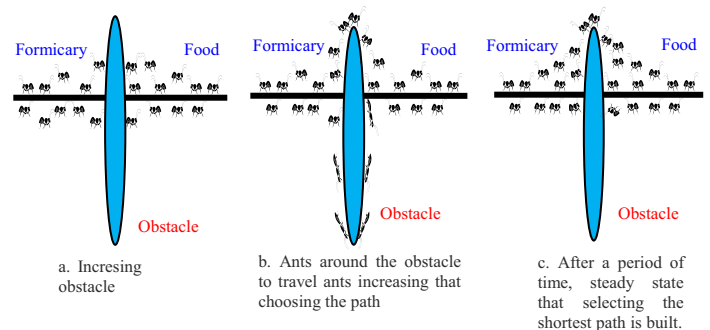


Figure 1 Schematic of ants foraging routing

2 GA Algorithm Overview

Genetic algorithm, abbreviated as GA), is a theory of evolution proposed by John Holland in 1975 firstly[3]. Genetic algorithm is a global optimization algorithm which is developed in recent years, which by selection, genetic variation and other mechanisms to achieve improving the adaptability of each individual. Genetic algorithms have been successfully used in a variety of learning tasks and optimization problems[4].

3 ACO-GA Model Description and Parameter Setting

Currently, air traffic control is mainly based on the distance intervals to ensure flight safety[5]. In order to achieve the automation system, the concept of time base[6] was introduced. In this paper, the minimum wake turbulence separation standards between different types of flights ICAO (ICAO), as shown in TABLE I.

TABLE I. Aircraft wake turbulence separation standard (s) ICAO provisions

	Wake	Back Aircraft		
		(H)	(M)	(L)
	(H)	99	114	167
	(M)	74	74	138

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Front Aircraft	(L)	74	74	98
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Therefore, the objective function is established in this paper:

$$\min DT = \sum_{i=1}^{M-1} w_{(i,i+1)} \quad (1)$$

Where, M : the number of flights that waiting for takes off; $w_{(i,i+1)}$: the wake between flight i and $i+1$

4 Algorithm Design of ACO-GA

Assuming that the number of flights which are waiting for takes off is M , the number of ants is M in this article.

4.1 Pheromone Update Strategy and The State Transition

Probabilities

The flight collection that is to be sorted is P , the connection between any two flights consisting of the arc set A of all flights. It is possible to build graph $G = (P, A)$, and G is full connected. The definition of pheromone and the state transition probabilities have important significance for the proposed algorithm[7].

- pheromone update strategy

In this article, selecting pheromone synchronized manner, that is, after M ants complete their traversal search, update pheromone and adjustments unified (to strengthen or volatile). Assuming $\tau_{ij}(n)$ is the amount of pheromone after $(n-1)$ cycles, then:

$$\tau_{ij}(n) = (1-\rho) \tau_{ij}(n-1) + \sum_{k=1}^M \Delta \tau_{ij}^k \quad (2)$$

Wherein, $\rho \in (0,1)$ is a pheromone evaporation, $(1-\rho)$ is the residual factor pheromone; $\Delta \tau_{ij}^k$ is the pheromone amount that k -th ant left in path (i,j) of this cycle. According to Marco Dorigo's Ant-Cycle model, we can get:

$$\Delta \tau_{ij} = \begin{cases} \frac{Q}{C_{min}}, & \text{if } (i,j) \text{ is an arc on the optimal path in this loop} \\ 0, & \text{else} \end{cases} \quad (3)$$

Where, Q is the pheromone intensity; C_{min} is the minimum time-consuming of departure of M ants in this cycle. This update method will not be able to guarantee unlimited accumulation of pheromone.

- State Transition Probabilities and Inspiration Information

Ant state transition probabilities for the proposed algorithm:

$$P_{ij}^k(t) = \begin{cases} \frac{[\tau_{ij}(t)]^\alpha [\eta_{ij}(t)]^\beta}{\sum_{k \in allowed_k} [\tau_{ik}(t)]^\alpha [\eta_{ik}(t)]^\beta} & j \in allowed_k \\ 0 & \text{else} \end{cases} \quad (4)$$

Formula (4), $\tau_{ij}(t)$ the amount of pheromone at time t .

for (i,j) . Inspiration information is $\eta_{ij}(t) = \begin{cases} \frac{1}{d_{ij}}, & i \neq j \\ 0, & i = j \end{cases}$,

d_{ij} represents the weights of (i,j) (wake turbulence of (i,j)).

$\eta_{ij}(t)$ is greater and makes a greater contribution to the state transition probabilities if the wake turbulence separation between i and j is smaller. p is a random number between 0 and 1. $allowed_k$ is the flights collection that ant k could select in next. α is the parameter of importance of pheromones; β is the parameter of importance of inspiration information.

4.2 Encoding of ACO-GA

Serial number encoding was used in this paper, this encoding is relatively simple. And assume that the number of flights to be sorted is 20 (the length of chromosome is 20). Randomly generated 20 flights to be sorted as follows: ($<0,M><1,H><2,H><3,L><4,M><5,L><6,H><7,M><8,M><9,H><10,H><11,M><12,M><13,L><14,M><15,M><16,M><17,L><18,H><19,H>$). Each group represents a gene wherein: numeric represents the flight code (number), the letters represent the type of flight.

4.3 Design of Mutation Operator

As the number coding is used in encoding, the simplest and most intuitive way is that swap the positions of the pair flights, resulting the sequence of flights of mutation. For example:

$G_x = (<0,M><1,H><2,H><3,L><4,M><5,L><6,H><7,M><8,M><9,H><10,H><11,M><12,M><13,L><14,M><15,M><16,M><17,L><18,H><19,H>)$.

We can obtained the new flight sequence G'_x after variation by exchanging position 3 and position 18 in flights sequence G_x , $G'_x = (<0,M><1,H><2,H><16,M><4,M><5,L><6,H><7,M><8,M><9,H><10,H><11,M><12,M><13,L><14,M><15,M><3,L><17,L><18,H><19,H>)$.

4.4 Selection of each parameter value of ACO-GA

The parameters are selected as shown in TABLE II according to the characteristics of the problem itself:

TABLE II. Simulation Parameters

	Parameters name			
	α	β	ρ	Q
Value	0.5	1.0	0.7	120

4.5 The flow of ACO-GA algorithm

Here are the concrete steps to solve the flights sorting by using the combination of the ant colony algorithm and genetic algorithm.

Step 1: The initial flights sequence $A = \{1,2,\dots,20\}$ is a flights collection to be sorted, $D = (d_{ij})_{20 \times 20}$ is the minimum safe separation matrix for the arbitrary pair flights (i,j) . Assuming loop times is $NC_{max} = 2000$, the number of ants is $M = 20$, the parameter of importance of pheromone $\alpha = 0.5$,

the parameter of importance of inspiration factor $\beta=1$,

$$\text{inspiration information } \eta_{ij} = \begin{cases} \frac{1}{d_{ij}}, i \neq j \\ 0, i = j \end{cases}, \text{ volatile}$$

coefficient $\rho = 0.7$, pheromone strength $Q = 120$. The initial pheromone of (i, j) which are adjoining flights to be sorted is $\tau_{ij} = 1$, $\Delta\tau_{ij} = 0$. The initial value of $C_{\max}$ is obtained by using the maximum wake of the minimum wake multiply the number of flights which to be sorted minus one; $\text{Tabu}(k)$ is a taboo table that represents the combination of flight sorting of ant k , the initial $\text{Tabu}(k)$ is an empty set, the flight collection that ant k could select is $\text{allowed}_k = \{A - \text{Tabu}(k)\}$, $1 \leq k \leq M$; assuming $NC = 0$;

Step 2: Ant k selects flight i as the first flight randomly;

Step 3: FOR $m=2$ to 20

FOR $k = 1$ to 20. When ant k has selected flight i ,

IF $\text{allowed}_k = \emptyset$, then the computing of k -th ant was completed;

ELSE the k -th ant selects back flight j from flight i with probability:

$$P_{ij}^k(t) = \begin{cases} \frac{[\tau_{ij}(t)]^\alpha [\eta_{ij}(t)]^\beta}{\sum_{k \in \text{allowed}_k} [\tau_{ik}(t)]^\alpha [\eta_{ik}(t)]^\beta} & j \in \text{allowed}_k \\ 0 & \text{else} \end{cases}$$

$\text{Tabu}(k) = \text{Tabu}(k) \cup \{j\}$, $i:=j$, updating allowed_k ;

END IF

END FOR

END FOR

Step 4: Checking $\text{Tabu}(k)$ of each ant, that the sorting portfolio of 20 flights, and calculating the total leave

time-consuming of these combinations. Locating the minimum leave time-consuming $C_{\min}(k)$ in this loop and corresponding sequence of flights G_x . if $C_{\min}(k) < C_{\max}$, then $C_{\max} = C_{\min}(k)$, else keeping $C_{\max}$ constant.

Step 5: Taking mutation operation for G_x relying on $C_{\min}(k)$ and G_x calculated by Step4. $C_m(k)$ is the total delay of flights sequence after mutation, compare $C_{\min}(k)$ and $C_m(k)$. If $C_{\min}(k)$ is smaller than $C_m(k)$, the mutation operation will not stop until $C_{\min}(k)$ is not smaller than $C_m(k)$. Updating minimum value of consuming time of departure $C_{\max} = C_{\min}(k)$, and the corresponding sequence of flights G'_x ;

Step 6: Updating pheromone. Taking reinforce operation to the adjacent flights (i, j) of G'_x in this iteration, otherwise reducing the pheromone;

IF $(i, j) \in G'_x$

$$\Delta\tau_{ij} = \Delta\tau_{ij} + \frac{Q}{C_{\max}};$$

$$\tau_{ij} = (1 - \rho) \bullet \tau_{ij} + \Delta\tau_{ij};$$

ELSE

$$\tau_{ij} = (1 - \rho) \bullet \tau_{ij};$$

END IF

Step 7: $NC = NC + 1$, if $NC < NC_{\max}$, clearing $\text{Tabu}(k)$, and go to Step1; otherwise breaking the loop, output the final result.

5 Simulation and Analysis of Experimental Results

The simulation results are shown in Table 3:

TABLE III. Comparison table about the total time of departure between before and after optimization

Flight No.	FCFS		ACO		ACO-GA	
	Random sequence(Q_0)	Time-consuming(s)	Optimized sequence(Q_1)	Time-consuming(s)	Optimized sequence(Q_2)	Time-consuming(s)
1	M(0)	0	M(0)	0	L(5)	0
2	H(1)	74	H(19)	74	L(3)	98
3	H(2)	173	H(18)	173	L(13)	196
4	L(3)	340	H(10)	272	L(17)	294
5	M(4)	414	H(9)	371	H(1)	368
6	L(5)	552	H(6)	470	M(12)	482
7	H(6)	626	H(2)	569	M(8)	556
8	M(7)	740	H(1)	668	M(0)	630
9	M(8)	814	M(16)	782	M(16)	704

10	H(9)	888	M(15)	856	M(14)	778
11	H(10)	987	M(14)	930	M(7)	852
12	M(11)	1101	M(12)	1004	M(11)	926
13	M(12)	1175	M(11)	1078	M(15)	1000
14	L(13)	1313	M(8)	1152	M(4)	1074
15	M(14)	1387	M(7)	1226	H(6)	1148
16	M(15)	1461	M(4)	1330	H(2)	1274
17	M(16)	1535	L(17)	1438	H(19)	1346
18	L(17)	1673	L(13)	1536	H(9)	1445
19	H(18)	1747	L(5)	1634	H(10)	1544
20	H(19)	1846	L(3)	1732	H(18)	1643

TABLEIII shows the analysis: ACO-GA flights sorting algorithms on optimization problems better than FCFS algorithm and ant colony algorithm and can effectively reduce the total time flight departure. Meanwhile, this paper presents

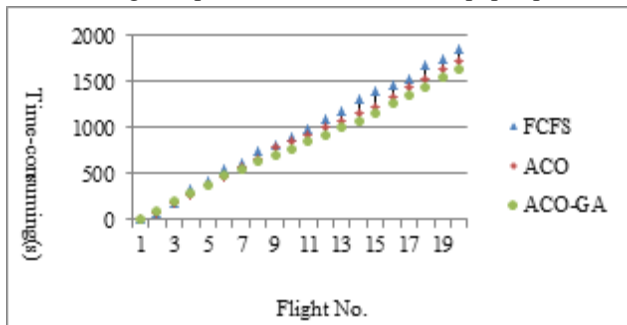


Figure 2. Price chart of flight number and departure time-consuming

As we can see from Figure 2, ACO-GA algorithm is superior to the ant colony algorithm and FCFS algorithm. Not only does figure 3 illustrates the iteration times to obtain the optimal sequence which using ACO-GA is less than ACO algorithm needed, but also illustrates the ACO-GA is superior to ACO.

6 Conclusion

This paper studies that research on the flight sorting by combining the ant colony optimization (ACO) and genetic algorithm (GA), and established a mathematical model of optimization, the model was solved finally. The experimental results showed that: compared with FCFS algorithm and ant colony algorithm, ACO-GA algorithm can effectively shorten the total flight time-consuming of departure, and can get the best flight departure sequence. Further studies for future may be considered that multi-track departure sequencing optimization problems and the practical application.

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price chart of flight number and departure time-consuming; Scatterplot of the times of iteration and the flight departure time-consuming, as shown in Figure 2, 3 below:

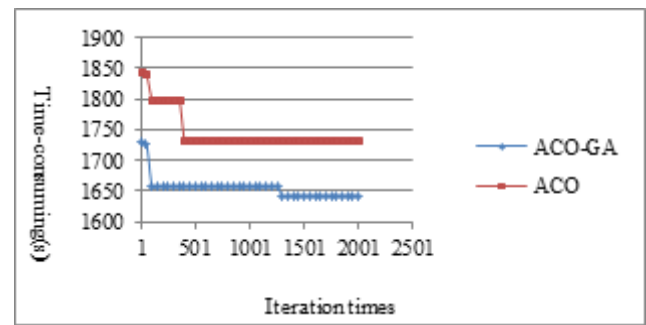


Figure 3 Scatterplot of the times of iteration and the flight departure time-consuming

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